ZentroMart VoiceOps — Final Evaluation

Capstone submission evidence | 23 October 2026

Scope

Synthetic-data customer-support agent with prompting-only baseline, policy-grounded RAG, guarded mock order tools, multi-turn routing, and live Gnani STT/TTS demonstration.

Recorded metrics

Measurement	Verified result
Dataset integrity	PASS: 9 CSVs, 21 KB documents
Pytest	23 / 23 passing
Offline decision routing	96 / 96 exact matches (RAG stub, no ticket side effects)
Baseline generated requests	51 LIVE; 1 HTTP error (52 evaluated)
RAG generated requests	52 LIVE (52 evaluated)
Mean gold-section Recall@5	64.89% (47 evaluable prompts)
Answer accuracy	Not independently adjudicated

Observed live voice integration

The user reported successful Gnani TTS audio generation, Gnani STT transcription, a complete STT → support agent → TTS voice answer, and an interactive multi-turn Streamlit conversation. API credentials, the WAV demonstrations and a video were not included in the final ZIP; live behavior is therefore user-reported, not independently reproduced during packaging.

Evaluation interpretation

The 96/96 figure measures exact offline action labels, not free-form answer quality or secure production ticket handling. Recall@5 is the mean fraction of expected KB section IDs found among the top five retrieved passages; it is neither an official project grade nor factual-answer accuracy. Baseline and RAG status counts measure completed requests only.

Known limitations

The baseline lacks policy/account context by design. Retrieval still misses some applicable sections and some generated answers require adjudication, particularly security escalations and high-value damaged products. Authentication and tickets are synthetic; session state is local; ticket creation is disabled in the interactive demonstration. The project lacks independently measured runtime/cost, a demo video and production deployment hardening.

Reproducibility

Use the unmodified dataset separately, install dependencies, run `python -m pytest -q` and `python scripts/evaluate_stage3_offline.py`. Re-running the full `scripts/evaluate.py` requires configured Groq credentials and quota. The included JSON reports preserve the last evaluation outputs.